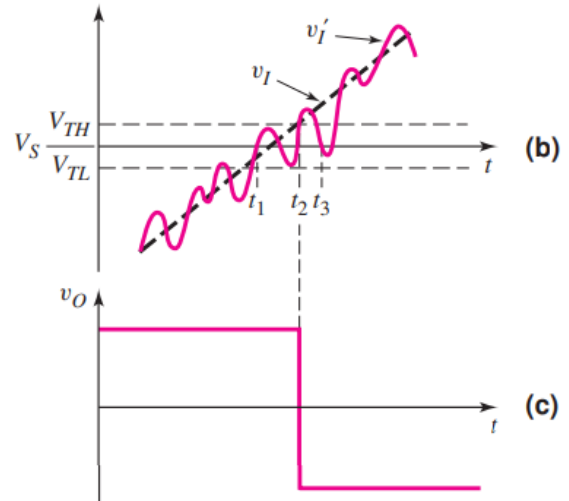
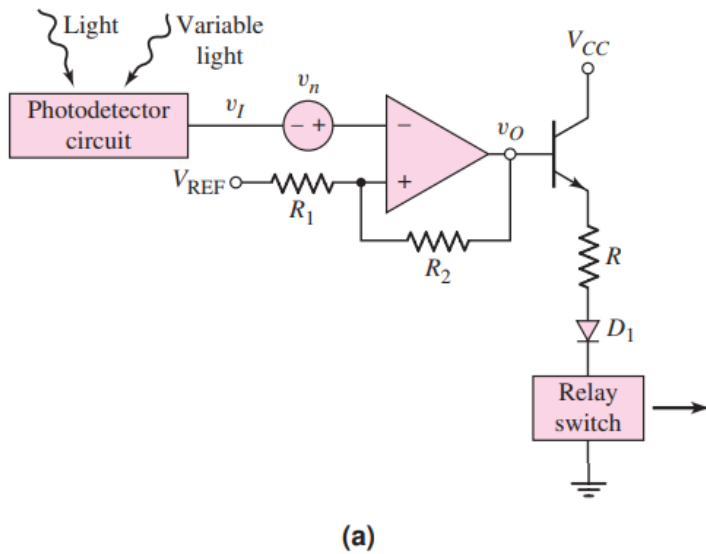


Exercise 1



Design the street light control circuit shown in the above figure such that the switching voltage is $V_S = 2\text{ V}$ and the hysteresis width is 200 mV . Assume

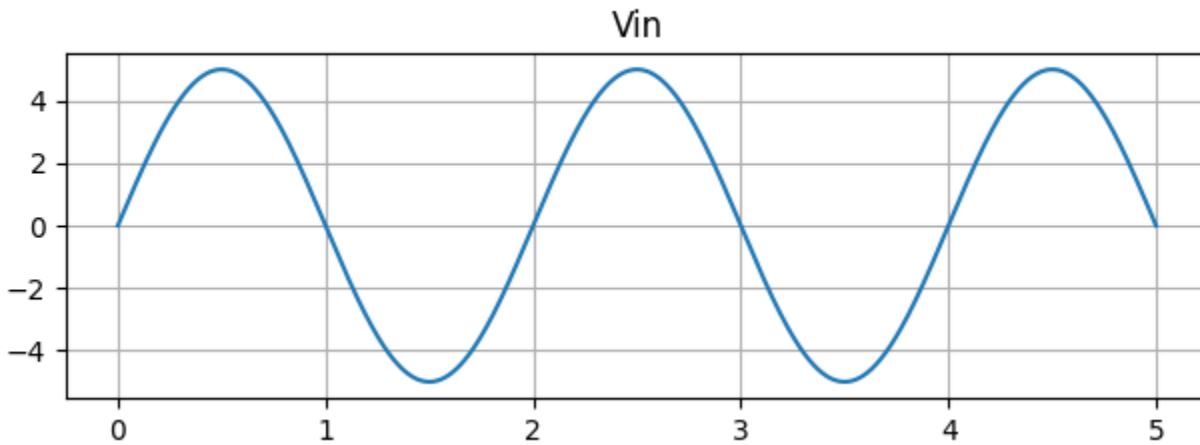
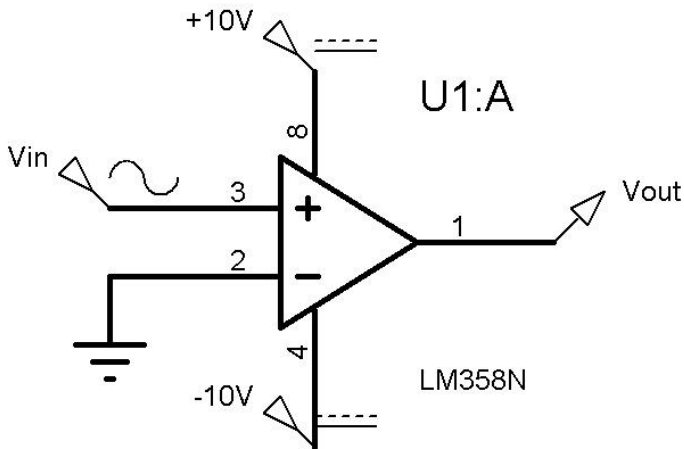
$$V_H = 15\text{ V} \text{ \& } V_L = -8\text{ V}.$$

- Find the resistances in $k\Omega$. You can assume the value of one resistance.
- Find the value of V_{REF} .
- Suppose, the current through the resistance R is $250\text{ }\mu\text{A}$ when $v_o = V_H$ and relay switch resistance is $330\text{ }\Omega$. Find the value of R .

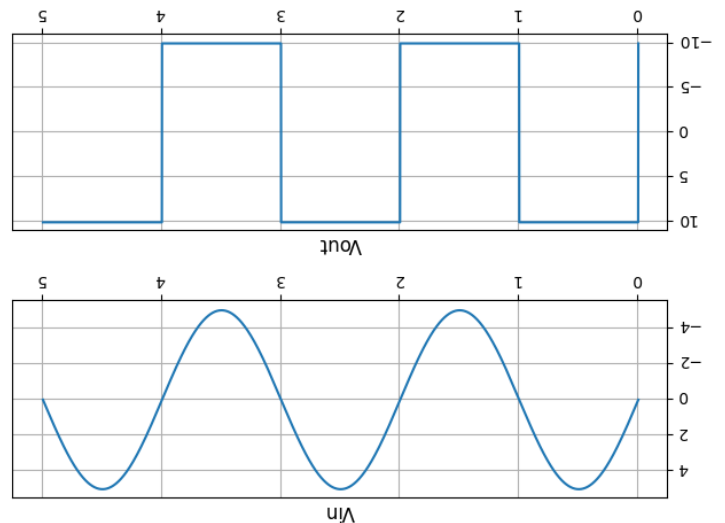
Ans: a) 10, 1140
b) 2.0175 V
c) 54.07 k Ω

Exercise 2

For the OP-AMP comparator circuit below, V_{in} vs time plot is given. Draw the V_{out} vs time plot for the given input.

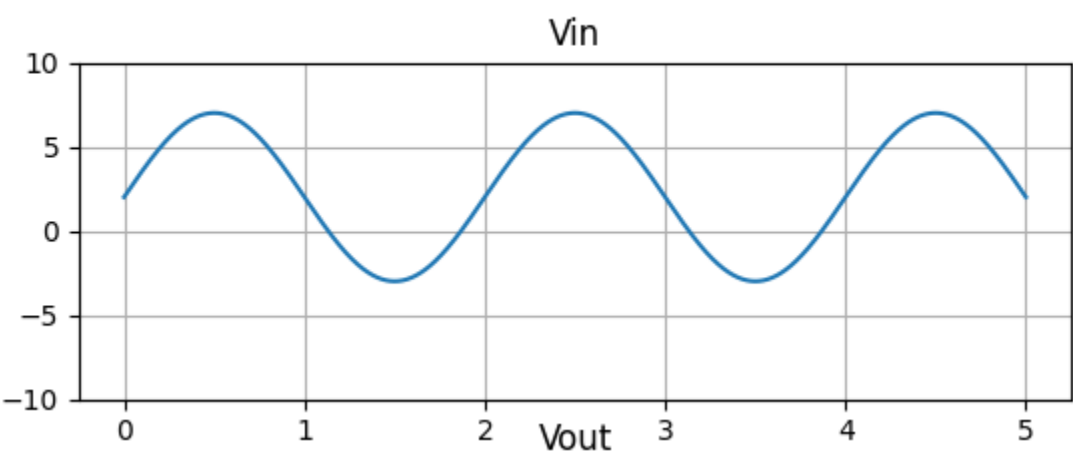


Ans:

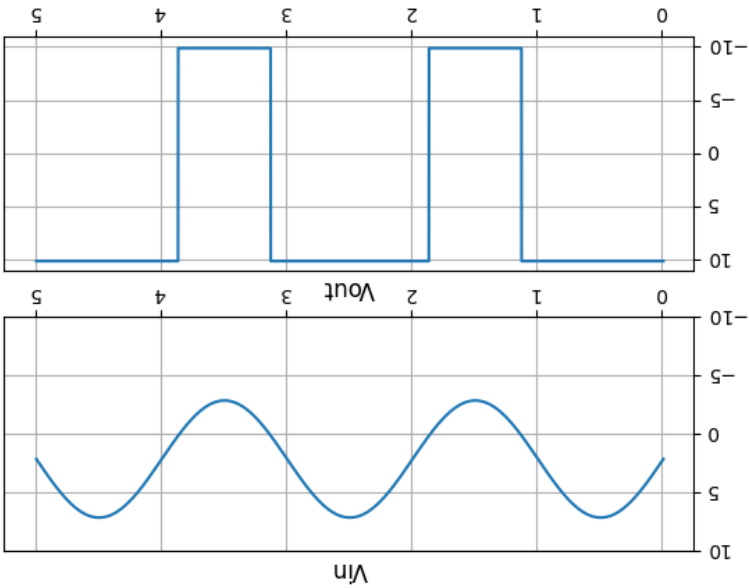


Practice Problem 1:

Do the same as in **Exercise 2** for the following:

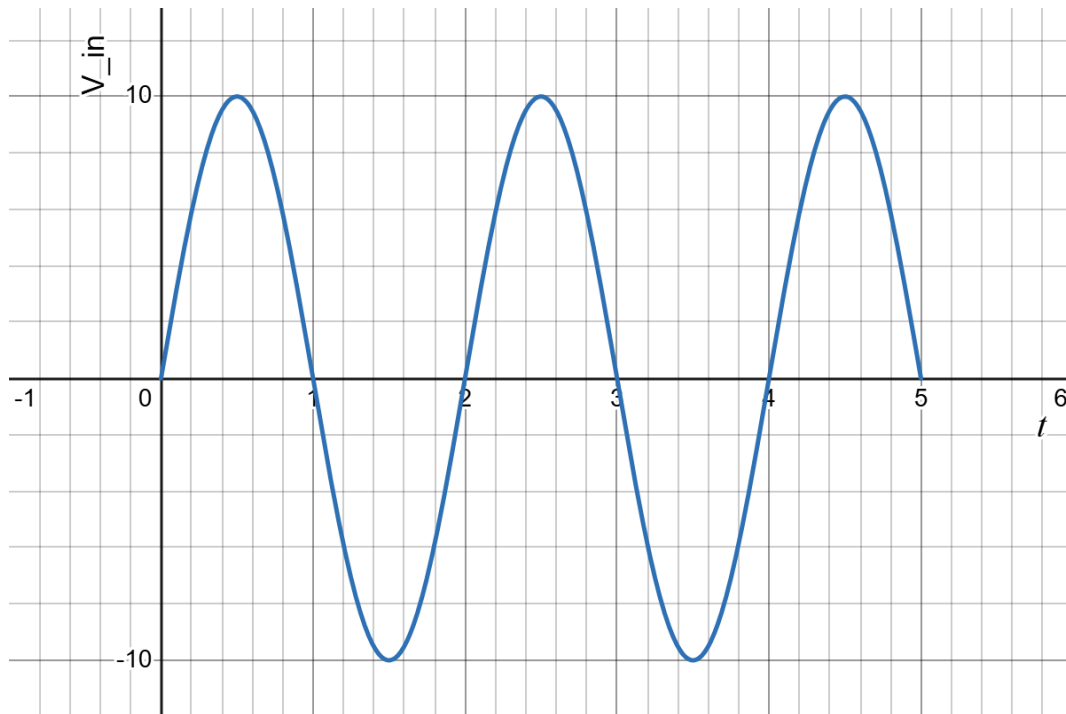
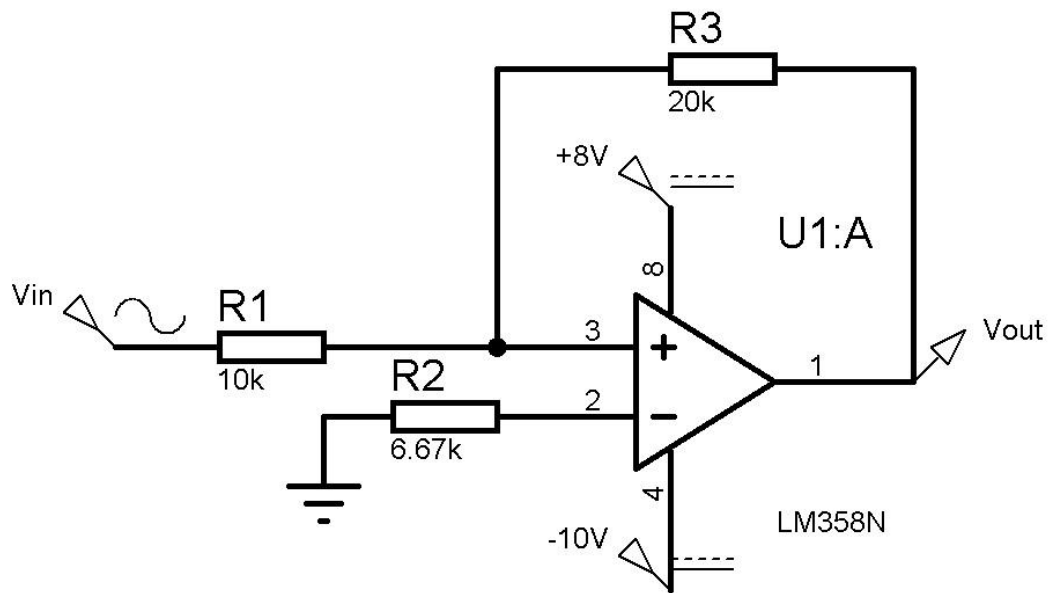


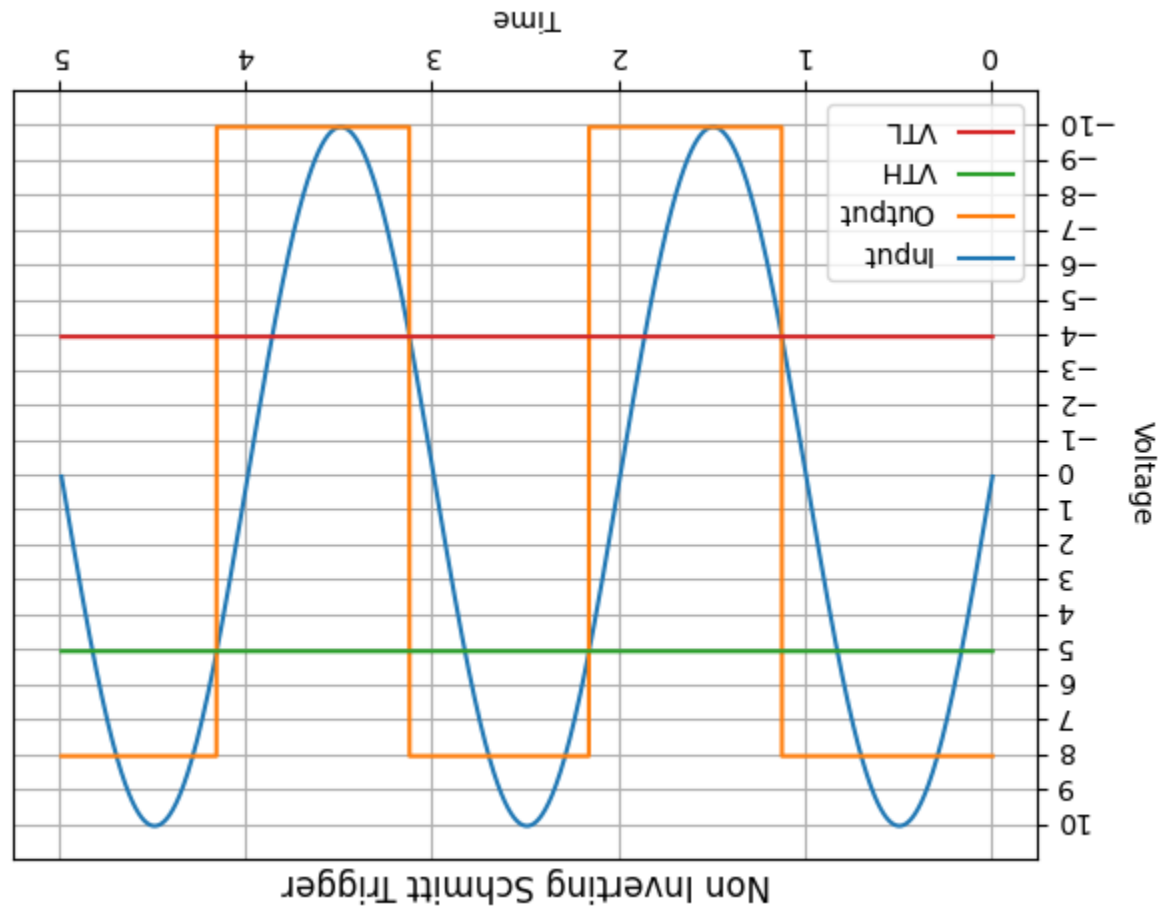
Ans:



Exercise 3

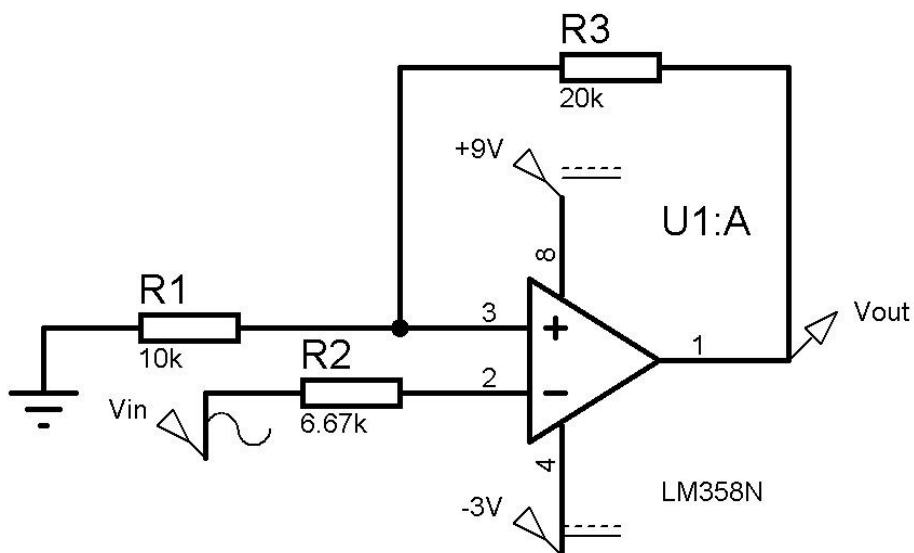
For the non-inverting Schmitt Trigger circuit below, V_{in} vs time plot is given. Draw the V_{out} vs time plot.

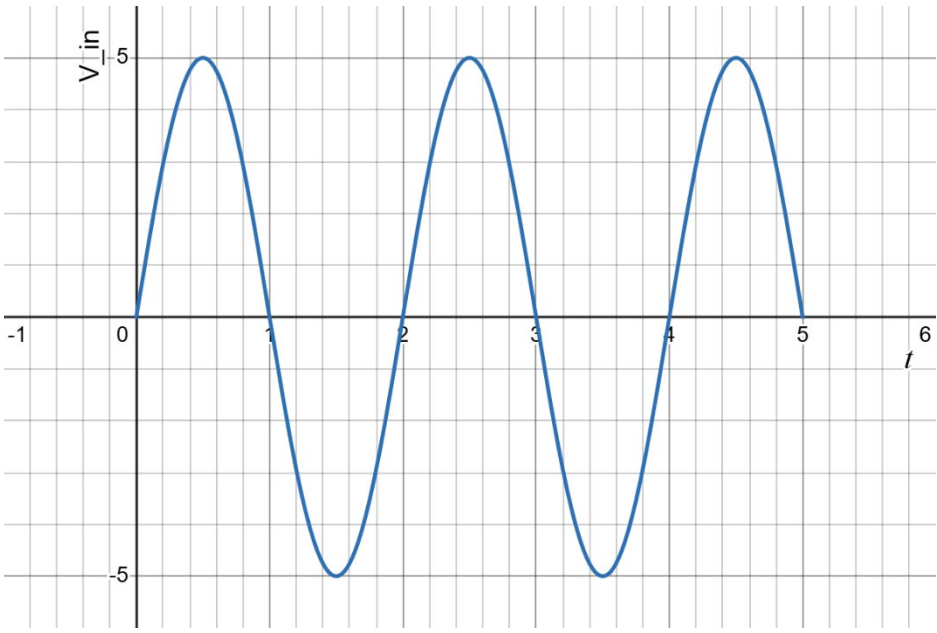




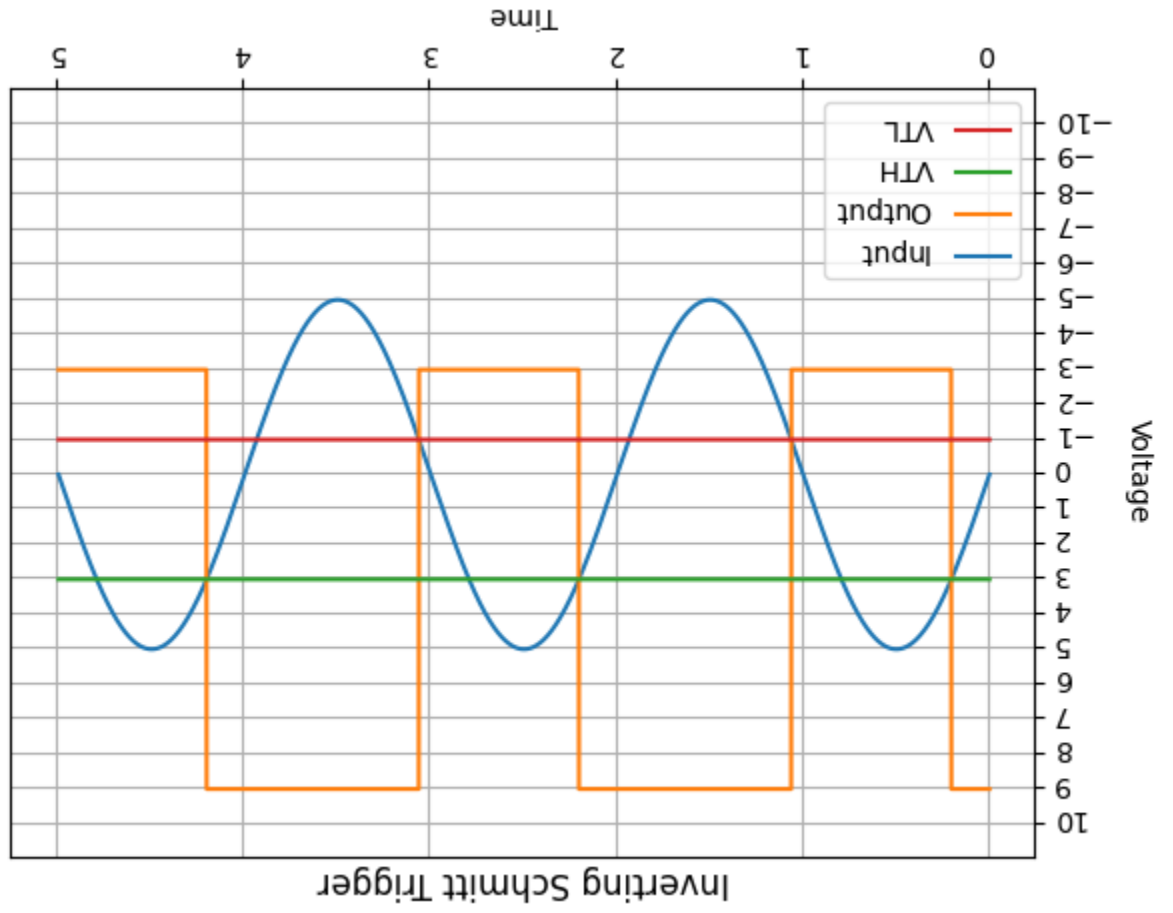
Exercise 4

For the inverting Schmitt Trigger circuit below, V_{in} vs time plot is given. Draw the V_{out} vs time plot for the given input.

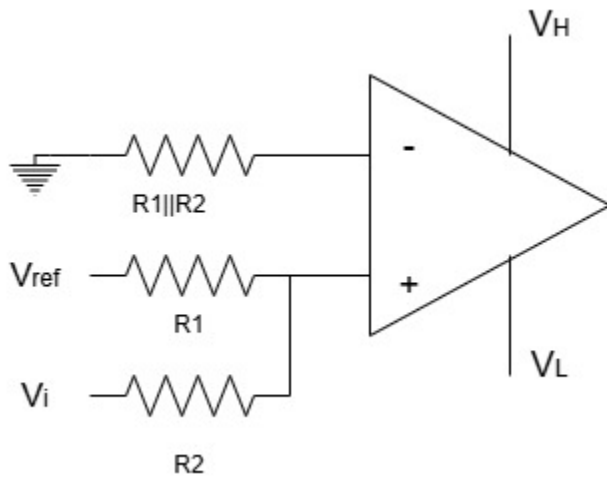




Ans:



Exercise 5

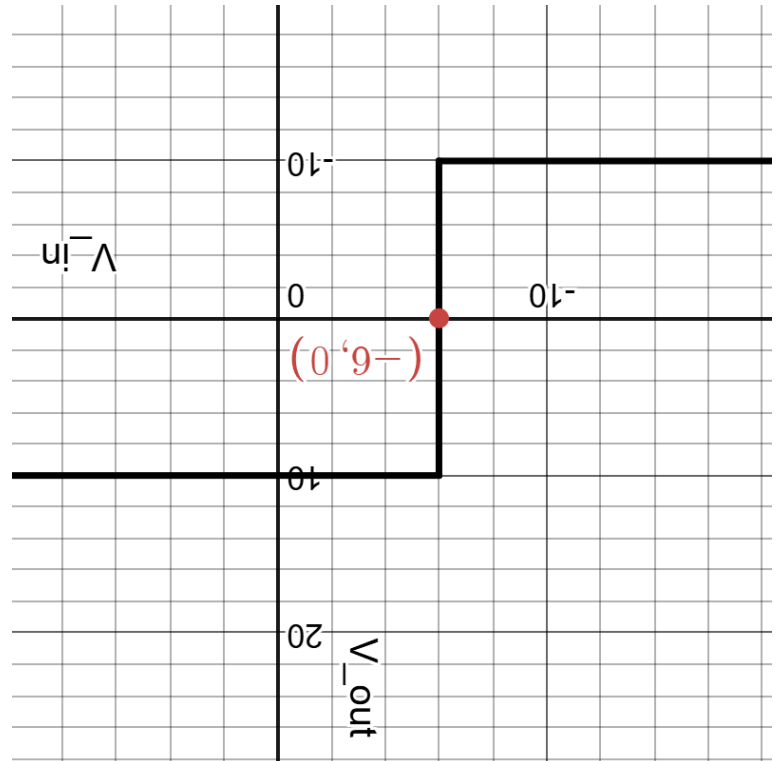


If $R_1 = 10\text{ k}\Omega$, $R_2 = 20\text{ k}\Omega$, $V_{ref} = 3\text{ V}$

$V_H = 10\text{ V}$, $V_L = -10\text{ V}$

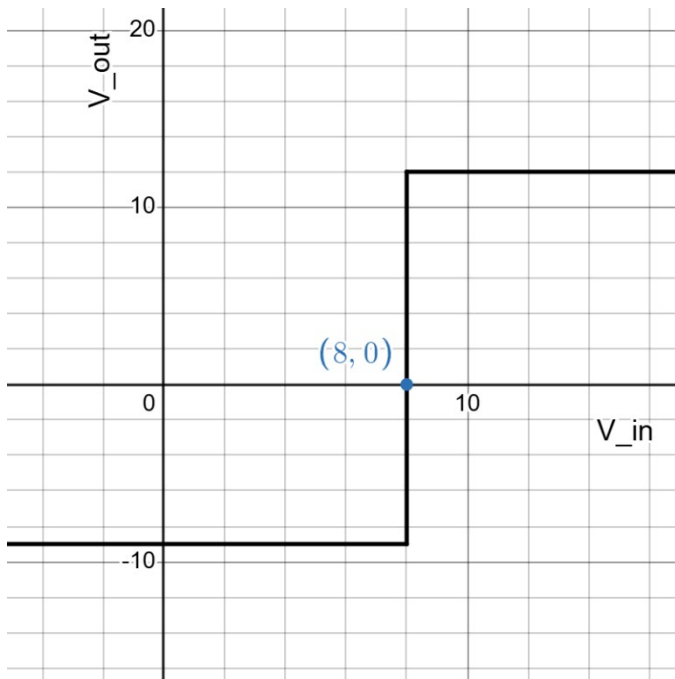
Draw the VTC of this circuit with proper labeling

Ans:



Exercise 6

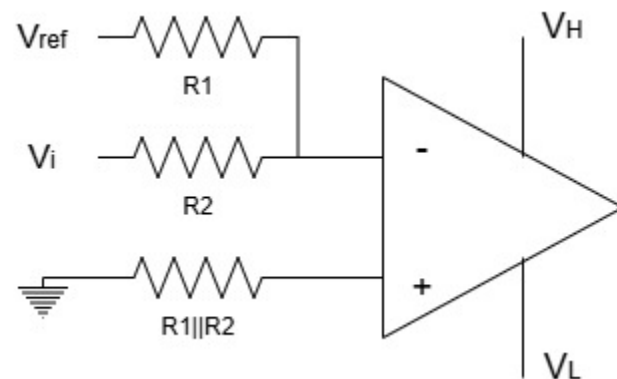
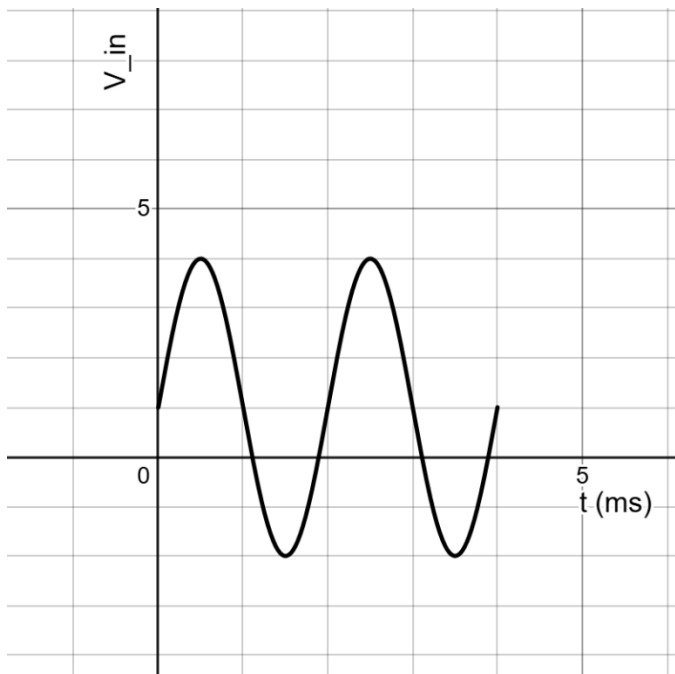
Design a comparator circuit with the following VTC:



$$\text{Ans: } R_1 = 5 \text{ k}\Omega, R_2 = 10 \text{ k}\Omega, V_{ref} = -4 \text{ V}, V_H = 11 \text{ V}, V_L = -9.5 \text{ V (Non-Inverting)}$$

Exercise 7

Draw the V_{out} vs time plot for the given V_{in} vs time plot to the following circuit:

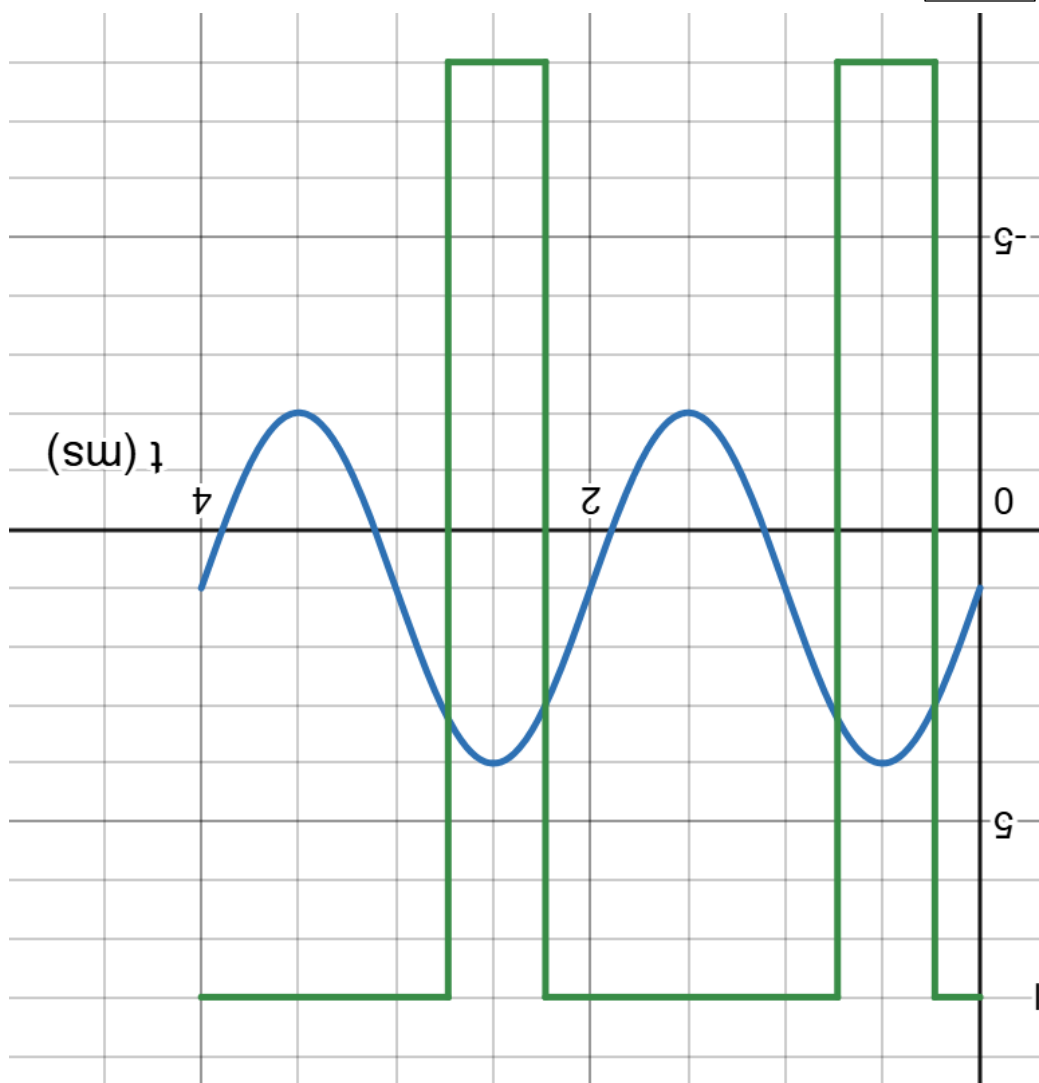


$$V_{ref} = -2 \text{ V}$$

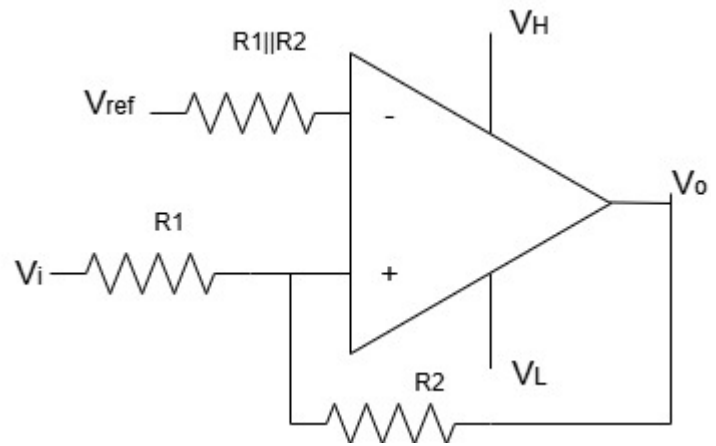
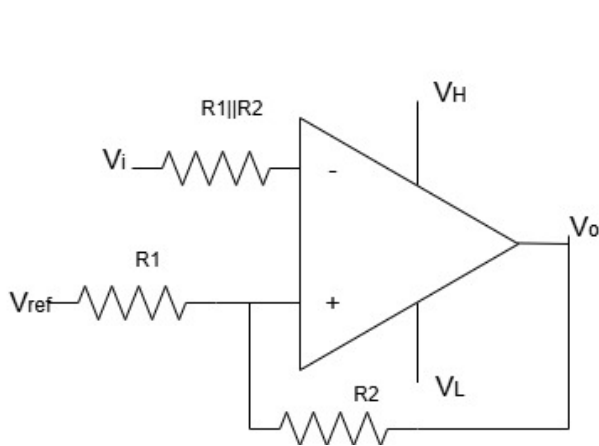
$$R_1 = 10 \text{ k}\Omega, R_2 = 15 \text{ k}\Omega$$

$$V_H = 8 \text{ V}, V_L = -8 \text{ V}$$

Ans:



Exercise 8

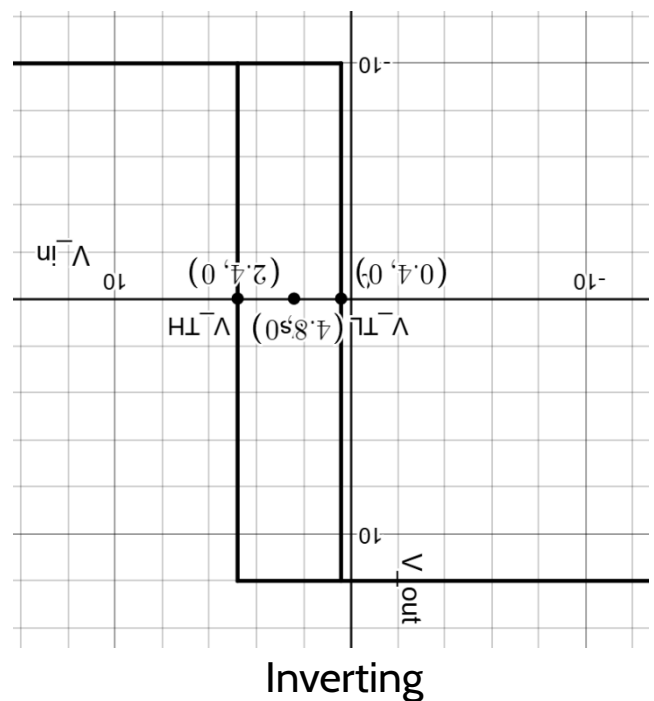
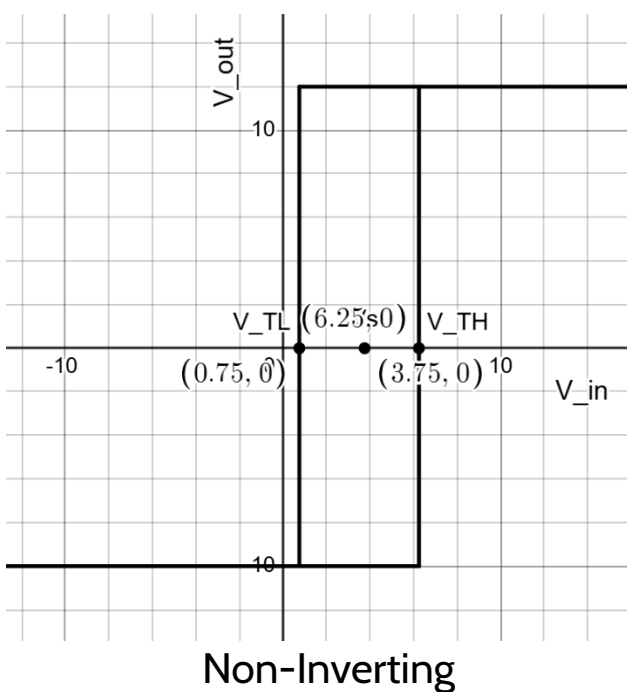


For the above two circuits,

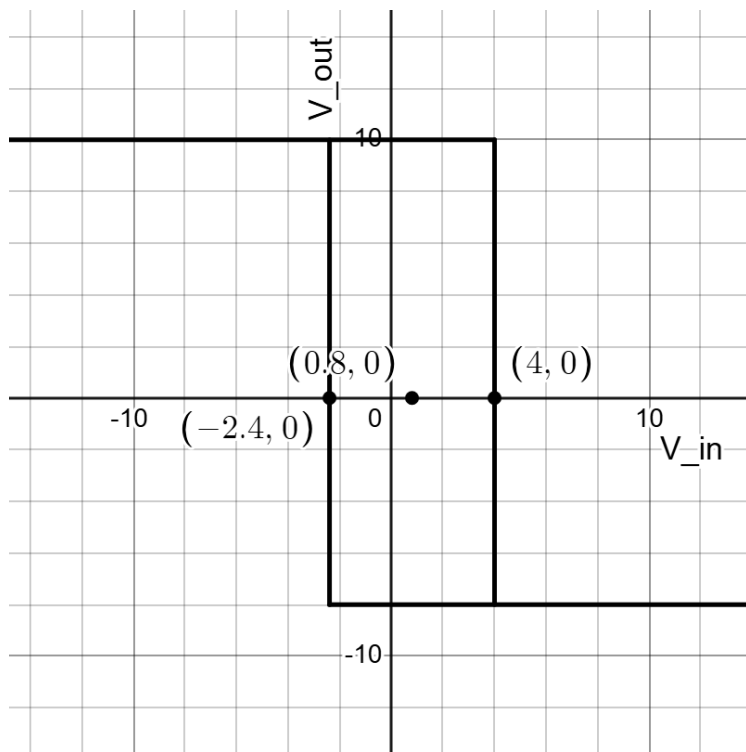
$$R_1 = 5 \text{ k}\Omega, R_2 = 20 \text{ k}\Omega, V_{\text{ref}} = 3 \text{ V}, V_H = 12 \text{ V}, V_L = -10 \text{ V}$$

- Find higher & lower threshold voltages.
- Find shift voltages & hysteresis widths.
- Draw their VTCs.

Ans: a) $V_{TH} = 4.8 \text{ V}$ (Inverting), 6.25 V (Non - Inverting),
 $V_{TL} = 0.4 \text{ V}$ (Inverting), 0.75 V (Non - Inverting)
 b) 2.4 V (Inverting), 3.75 V (Non - Inverting)



Exercise 9



If $R_1 = 2 \text{ k}\Omega$, design a Schmitt Trigger to achieve the given VTC.

$$\text{Ans: } R_2 = 3.625 \text{ k}\Omega, V_{ref} = 0.68 \text{ V}, V_H = 10 \text{ V}, V_L = -8 \text{ V}$$